DATA 780 Group Project: Waterflow
Midway Report

Spencer Childress
School of Data and Information Sciences
University of North Carolina at Chapel Hill
Chapel Hill, NC
spncr@unc.edu

Jay Patel
School of Data and Information Sciences
University of North Carolina at Chapel Hill
Chapel Hill, NC
jmjay1@email.unc.edu

Ryan Willis
School of Data and Information Sciences
University of North Carolina at Chapel Hill
Chapel Hill, NC
rywillis@unc.edu

1 Introduction

We intend to build a machine learning model that predicts water flow in water basins without sensor data. The inputs to the model include land, soil and climate attributes, which will be evaluated against water flow measurements from water basins with sensor data. Most water basins have no sensor data yet water managers need flow estimates for flood and drought planning. An inexpensive, interpretable method that can be easily deployed where we have confidence in its predictive power is useful, and understanding where the method fails improves our understanding of hydrologic processes.

Today, hydrologists tune a physical model for each river with historical measure water flow. For rivers without sensor data, they compare with nearby, similar "donor" rivers, which has proven to be failure-prone. Deep learning models trained on hundreds of rivers now do better [Kratzert et al., 2019], but they're computationally intensive and suffer from interpretability.

Our approach will test whether the standard practice that "similar rivers behave alike" improves with unsupervised water basin clustering and predictive models accounting for spatial proximity, weather patterns, and ecological attributes. We will compare models with minimal inputs, a basin's cluster, and its full attributes to determine the most accurate approach. We want to understand the most predictive characteristics, in particular whether basin clustering is an informative exercise.

Clustering may discard important information from the raw attributes of the water basins in the training data and predefined weather windows may need additional context to account for slow snowmelt and groundwater levels. However, clustering may help when training basins are scarce, where informatively characterizing a water basin from its raw attributes is impractical. Therefore, our main focus will be to evaluate whether clustering helps when training basins are scarce.

We will assess our model on its predictive accuracy of water flow, as measured with the Nash-Sutcliffe Efficiency metric:

$$\text{NSE} = 1 - \frac{\sum_t \left(Q_{\text{obs},t} - Q_{\text{pred},t}\right)^2}{\sum_t \left(Q_{\text{obs},t} - \bar{Q}_{\text{obs}}\right)^2}$$

We will train separate neural networks for each cluster. New and test observations will be assigned to the nearest cluster centroid and then routed to the corresponding neural network for prediction. To

Preprint. Work in progress.

avoid leakage we will split by basin using spatially-blocked folds; refit scalers PCA and clustering inside each fold; and train on earlier years to avoid temporal correlation.

2 Related Work

Previous research in this area for determining streamflow has used several different models and traditionally they have been process based models. These models are typically based on mathematical formulas that are common in hydrological sciences but there is room for improvement as they are by no means 100% accurate. In our main research paper reference (Kratzert et al., 2019) it is pointed out that machine learning methods have come a long way and being data driven they take into account more than the process based methods can. This is particularly useful for hydrologically diverse catchments where process based models typically need a deeper knowledge of a particular area which can be expensive/time consuming.

In the past Machine Learning models have been used to measure streamflow in ungauged basins even better than conceptual models that were calibrated for each catchment. This is even against gauged basins that are operating on a calibrated process model. While in the paper we researched this was on an average basis (didn't outperform all process/calibrated models), it does provide evidence that data based models are effective. This is particularly important for ungauged basins since even if financially hydrologists are able to start gauging the basin it will take 2-3 years to get a reliable data set.

Previous machine learning models researched were a type of recurrent neural network (RNN). They were specifically called Long short-term memory (LSTM) networks. These were used because they are great for long term models that look at weather effects that happen over time. They are useful for things like snowfall accumulation and vegetation growth. These happen over long lengths of time so it makes them useful for modeling basin catchment. In the results of their study they observed that LSTM network models using static catchment attributes was determined to be the most useful. We will be using this add a reference to see if we can create a machine learning model that is easier to interpret as well as less intensive.

3 Methods

The dataset will be derived from the Caravan v1.6 CSV release, specifically looking at US basins (camels), Great Britain basins (camelsgb), and Australia basins (camelsaus), totaling 1760 basins. Measurments for all basins will be daily from 1981-2014. Basins will be kept for analysis based on overall quality and completeness.

Static attributes for the kept basins will represent the physical and environmental characteristics of each basin. These attributes will be standardized prior to clustering and model development, with all preprocessing parameters estimated exclusively from the training catchments within each validation fold.

To prevent information leakage, catchment attributes derived directly from observed streamflow or other discharge-related measurements will be excluded from the predictor set. The remaining static catchment attributes will be examined for missing values, inconsistencies, and potential redundancy before model development.

Meteorological predictors will be constructed using fixed time intervals of 7, 30, 90, and 365 days. These intervals will capture short-term and longer-term hydrological and climatic conditions that may influence streamflow behavior, all of which will be applied consistently across all models.

The target variable, daily streamflow, will be log-transformed to reduce the influence of highly skewed flow distributions and improve model stability. Predictions will subsequently be evaluated in both transformed and original flow spaces to assess the implications of the transformation and back-transformation procedure.

Prior to model fitting, basins will be split into training and test sets. All preprocessing operations, including feature scaling and any dimensionality reduction, will be fitted independently within each training fold. This procedure will ensure that information from held-out catchments does not influence model development.

The scope of this analysis will involve the usage of several machine learning methods, both supervised and unsupervised. Ridge regression will serve as a baseline supervised method, and it will be run on lagged weather measurements and accuracy of nearest-gauge transfer.

After establishing a baseline ridge regression model, k-means clustering will be applied to the static catchment attributes to identify groups of physically similar river basins. Catchments will be classified into groups based on their similarity across the selected standardized attributes. The number of clusters (k) will be evaluated across three to four candidate values. This will allow the study to examine whether the granularity of catchment groupings influences predictive performance. Once clustering is done, each catchment will be assigned a cluster identifier. This identifier will subsequently serve as an alternative representation of catchment characteristics in the supervised methods.

An important consideration is that cluster membership is derived directly from the underlying catchment attributes. Consequently, cluster identifiers do not introduce additional information beyond the original attributes. Instead, they provide a potentially more parsimonious representation of catchment similarity that may improve predictive performance when the available training sample is small. The hydrological coherence of the resulting clusters will also be examined to determine whether catchments grouped together based on physical characteristics demonstrate meaningful similarities in their observed streamflow behavior.

Light Gradient Boosting Machine (LightGBM) will serve as the primary supervised learning method for streamflow prediction. LightGBM is a gradient-boosted decision tree framework that can accommodate nonlinear relationships and interactions between meteorological and catchment characteristics. For this model, 3 different configurations will be compared to determine best performance:

- No catchment information: The model will use meteorological forcing variables without static catchment attributes or cluster identifiers.
- Cluster-based information: The model will incorporate meteorological forcing variables and the corresponding unsupervised cluster identifier derived from the k-means clustering model.
- Raw catchment attributes: The model will incorporate meteorological forcing variables and the complete set of selected static catchment attributes prior to log transformation.

4 Preliminary Results

To preface, we have only begun exploratory data analysis following the finalization of our analysis plan, updated with the feasibility review from Professor Chiang. With the decision made to exclude the water basins out of Great Britain, our training and test data will be drawn from the following regions:

- `camels`, a database of 671 US water basins, of which 631 contain sufficient data
- `camelsaus`, a database of 561 Australian water basins, of which 541 contain sufficient data

Below is a map of the basins available in the two databases.

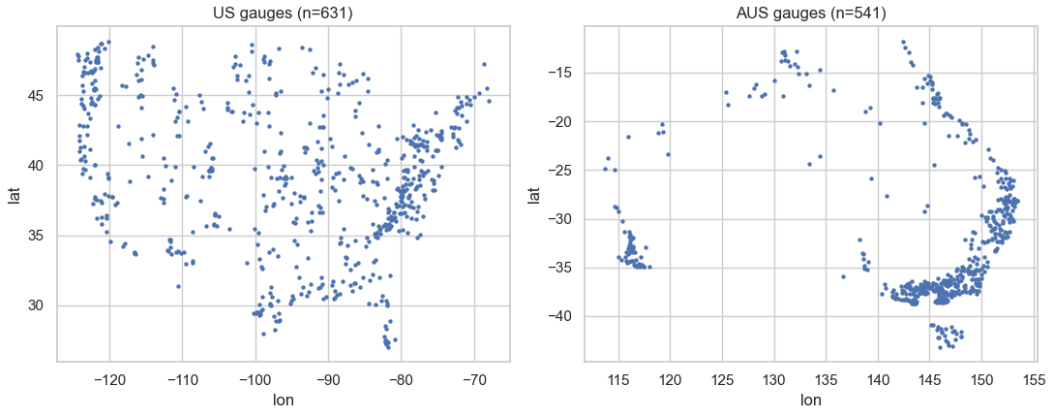


Figure 1: Map of the basins available in the `camels` and `camelsaus` databases.

To arrive at the set of water basins with sufficient data we applied the following thresholds:

- At least 15 years of valid water flow data.
- At most 15% of missing window days.

The following figures provide comparisons of the two databases.

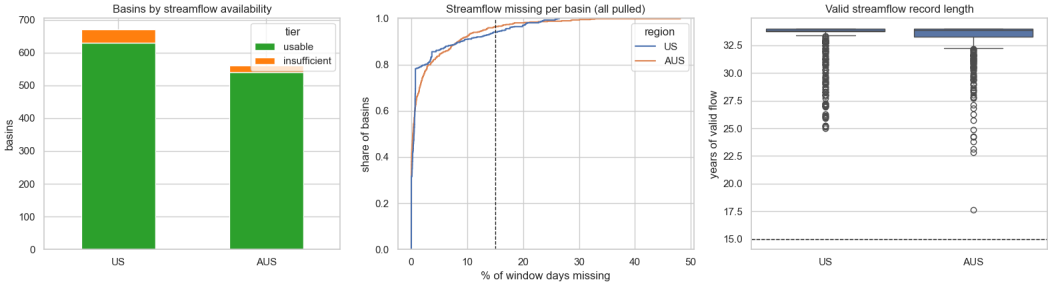


Figure 2: Comparison of the `camels` and `camelsaus` databases.

Below we dril down into the availability of streamflow data over time, from the early 1980s to the mid 2010s.

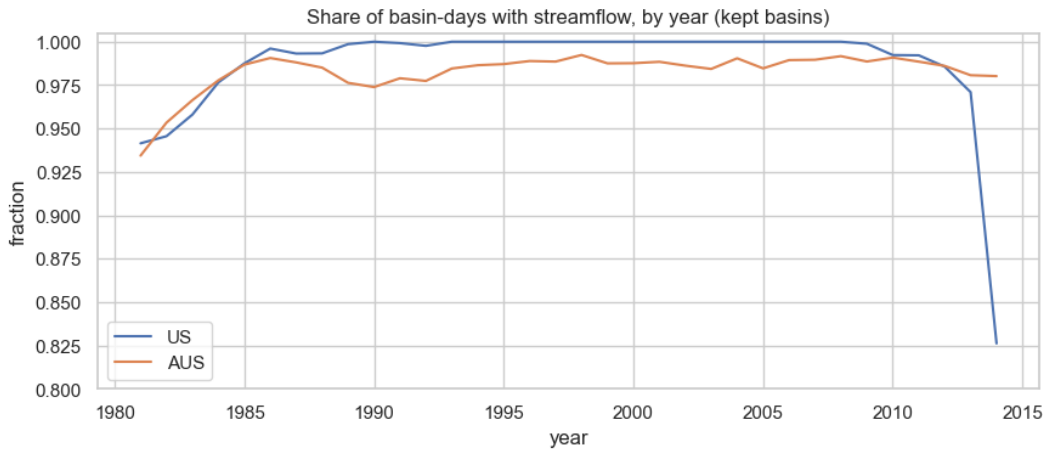


Figure 3: Availability of streamflow data by year.

To get an idea of the data present for each basin, we zoom in below on a single basin.

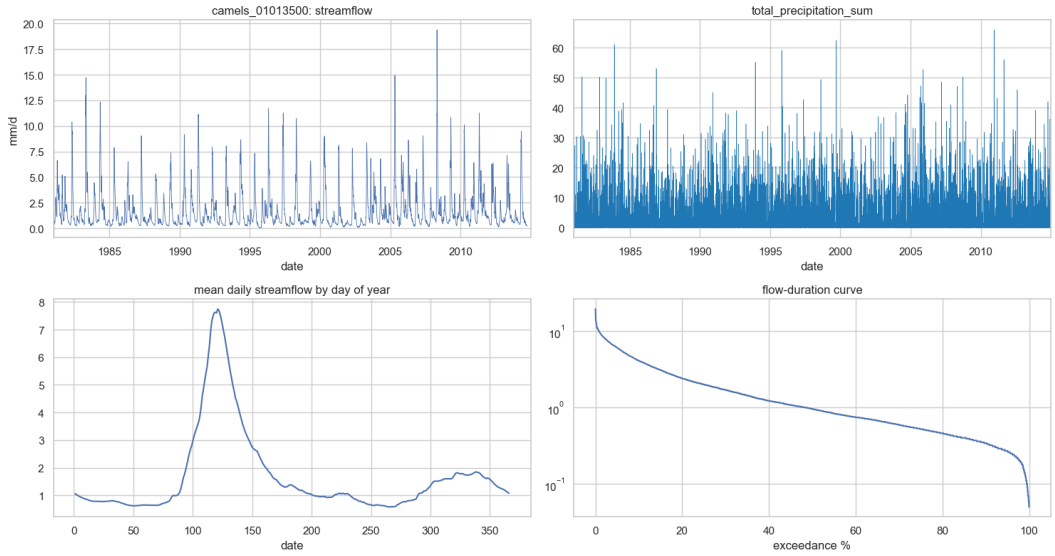


Figure 4: Streamflow data for a single example basin.

Note the seasonal spikes ˜120 days into the year where waterflow rises following spring precipitation. Our immediate next steps are to define the temporal split of our training and test data, and the locational blocking of our water basins, to inform k-fold validation.

5 Future Plans

The remaining work will focus on the leakage-safe US/AUS comparison and whether cluster information helps when training basins are scarce. The proposed milestones and division of labor are:

- **November 15 — Scope and evaluation protocol:** Spencer leads scope and validation; Jay leads data and features; Ryan supports clustering and evaluation.
- **November 22 — Core experiments and results freeze:** Ryan leads LightGBM runs; Jay supports data and baselines; Spencer leads the basin-scarcity analysis.
- **November 27 — Results synthesis:** Spencer leads interpretation; Ryan provides model diagnostics; Jay verifies data and forcing caveats.
- **November 29 — Full paper draft:** Spencer integrates the paper; Jay drafts data and methods; Ryan drafts modeling and results.
- **December 3 — Revision and reproducibility:** Spencer leads revisions; Jay verifies data claims; Ryan runs the clean-environment check.
- **December 5 — Final quality checks and presentation prep:** All review the paper and submission files; Ryan leads slide preparation; Spencer coordinates.
- **December 6 — Final report submission:** Spencer coordinates; all confirm the final report and accompanying files.
- **December 7 — Presentation:** Ryan leads the presentation; all teammates present and respond to questions.

Optional work (synthetic benchmark, CAMELS-GB, cross-region transfer, kernel regression, or expanded SHAP) will not delay the November 22 freeze; retain only a minimal synthetic positive control if the course requires it and time permits.

References

Frederik Kratzert, Daniel Klotz, Guy Shalev, Günter Klambauer, Sepp Hochreiter, and Grey Nearing. Toward improved predictions in ungauged basins: Exploiting the power of machine learning. *Water Resources Research*, 55(12):11344–11354, 2019. doi: 10.1029/2019WR026065.